2020 AMC 10B Problem 11 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10B Problem 11. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10B Problem 11

Problem:

Ms. Carr asks her students to read any 5 of the 10 books on a reading list. Harold randomly selects 5 books from this list, and Betty does the same. What is the probability that there are exactly 2 books that they both select?

Answer Choices:

A. $\frac{1}{8}$

B. $\frac{5}{36}$

C. $\frac{14}{45}$

D. $\frac{25}{63}$

E. $\frac{1}{2}$

Solution:

We don't care about which books Harold selects. We just care that Betty picks 2 books from Harold's list and 3 that aren't on Harold's list. The total amount of combinations of books that Betty can select is

$$\binom{10}{5} = 252$$

There are

$$\binom{5}{2} = 10$$

ways for Betty to choose 2 of the books that are on Harold's list. From the remaining 5 books that aren't on Harold's list, there are

$$\binom{5}{3} = 10$$

ways to choose 3 of them.

$$\frac{10 \cdot 10}{252} = \text{(D)}\boxed{\frac{25}{63}}$$

$$\textbf{OR}$$

We can analyze this as two containers with 10 balls each, with the two people grabbing 5 balls each. First, we need to find the probability of two of the balls being the same among five:

$$\frac{1}{3} \cdot \frac{4}{9} \cdot \frac{3}{8} \cdot \frac{5}{7} \cdot \frac{2}{4}$$

After that, we must multiply this probability by

$$\binom{5}{2}$$

for choosing the 2 balls that are the same chosen among 5 balls. The answer will be

$$\frac{5}{126} \cdot 10 = \text{(D)} \boxed{\frac{25}{63}}$$

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